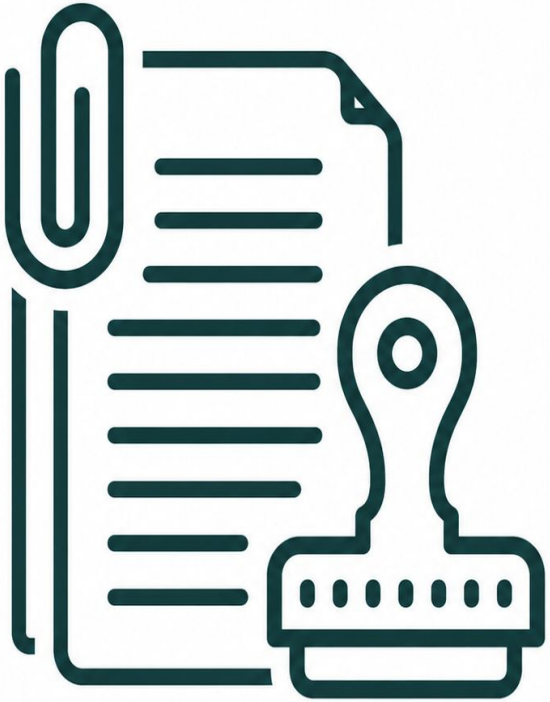
RCEL506
04/29/2026

DocTrace



Teaching Documents
to Speak

Presented by Pavan Sastry - Engineer I

Problem

Regulatory Documentation Efficiency

70%

of compliance team time spent on documentation

** Avatier Industry Report*

2–5%

manual error rate, driving audit risk & rework

** Industry QA benchmarks*

70–80%

time reduction potential with NLP automation

** Automated Publishing Study*

10 hrs → 2–3 hrs

A 100-document submission package that takes 10 hours manually is reduced to 2–3 hours with automation.

5–8 hrs per page

High-complexity regulatory documentation (SOPs, IND submissions) averages 5–8 hours per page manually. Process documentation can take weeks to months.

Automated publishing comparative study (Avatier, 2024)

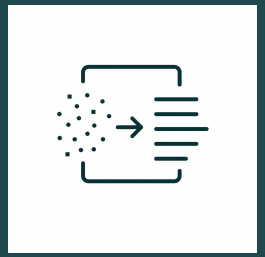
The Solution

An NLP Pipeline Trained on Legal Text

01

Raw Text

25k legal case passages



02

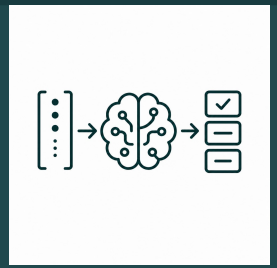
Tokenization and Noise Removal

Stop words removed, tokens extracted

03

TF-IDF Vectorization

Words have weighted numeric features



04

Classifier and Predict

LR or Random Forest predicts the label

NLP is a branch of AI that enables computers to understand, interpret, and generate human language.

The Justification

Why Legal Text?

Regulatory datasets are:

- Proprietary and confidential
- Not publicly available for model training

Legal datasets are:

- Large scale and labeled
- Publicly accessible

Shared Structure

Legal Documents:

- Over 50% of U.S. court opinions contain citations to prior cases

Regulatory documents:

- FDA and ISO standards require references to specific guidelines like SOPs ad ITRs

Shared Function

Regulatory compliance requires:

- Document traceability and auditability

Legal reasoning requires:

- Justification through citations

Both systems require::

- Explanations through evidence

Shared Data Characteristics

80-90% of enterprise data is unstructured text

Professionals spend:

- Up to 50% of their time searching for information

FDA Quality System Regulation (21 CFR Part 820) ;Unstructured data research (IDC/EMC; summarized in Wikipedia)

Value Proposition

The Financials

Company Size:		Annual Savings
Small	→	$32,000
Mid-size	→	$87,000
Large	→	$245,000

Griffith, C. (2024). “The ROI of Compliance Automation: Real Numbers from 100+ Companies.” FileFlo.

Model Results:

Metric	Random Forest (Best Model)	Naive Baseline
Accuracy	63% (+10%)	53%
Precision (Weighted)	65% (+37%)	23%
Recall (Weighted)	63% (+10%)	53%
F1 Score (Weighted)	57% (+20%)	37%

n ≈ 20,000 documents
Top 5 classes only, 10 fold cross validation
Stratified 80/20 train-test split

Business Justification

Per 1000 documents, DocTrace produces 100 more correct classifications than the baseline. That is 100 fewer misrouted documents.

Engineering Decision: This legal dataset is the best available public proxy for a regulatory text. This model is a feasibility proof, it demonstrates that NLP can learn treatment classification from formal document structure. Retraining on domain specific data is the next step, not necessarily a flaw.

⚠️ Domain shift:

Need to retrain and fine tune on regulatory SOP/ISO datasets to tailor it to each specific company

⚠️ Minority class recall:

Class weight tuning needed for applied and considered

⚠️ Vocabulary gap:

Regulatory terms may not be in the learned TF-IDF vocabulary

Conclusion

NLP can learn to classify how documents treat prior work.

- ✓ Random Forest achieved 63% accuracy – +10pp over the naive baseline
- ✓ TF-IDF successfully converts legal language into meaningful numeric features
- ✓ 10-fold cross-validation confirms the model is stable and generalizable
- → Next step: retrain on regulatory SOP datasets with domain-specific labels
- → Address minority class recall with SMOTE or cost-sensitive learning

Thank you

DocTrace
Documents Decoded